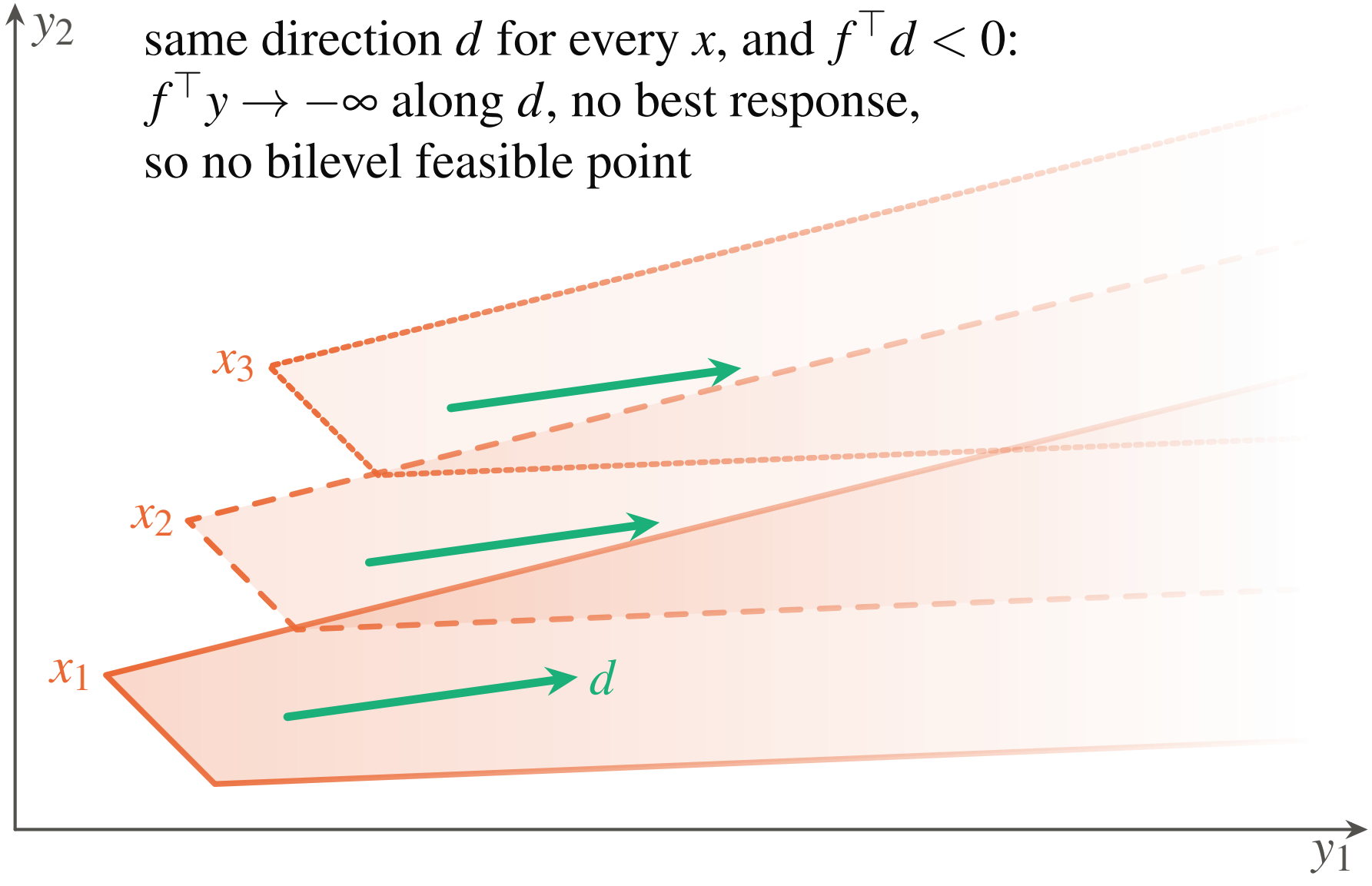
same direction d for every x, and $f^\top d < 0$:
$f^\top y \to -\infty$ along d, no best response,
so no bilevel feasible point
y_2
y_1
x_3
x_2
x_1
d